Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems

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Outline

Accuracy-Stability Trade-off

What is Forecast Stability?

Horizontal stability

- Measures how forecasts change across *different future time point* from the *same origin.*

Vertical stability

- Measures how forecasts change for the *same future time point* when the model is re-estimated at *different origins.*

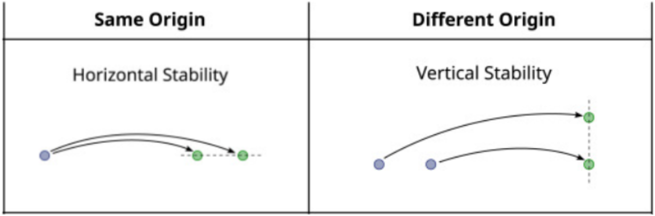


Figure 1: Godahewa, R., et al. (2025). *On forecast stability.*

Why it matters?

Forecasting drives operational decisions (inventory planning, workforce scheduling, budgeting and supply chain decisions, etc.)

Unstable forecasts lead to:

- revisions to plans and decisions,
- planning inefficiencies and higher costs,
- loss of credibility and trust in the forecasting system.

Why can a trade-off arise?

- Forecasting models are updated frequently.
- More frequent retraining $\Rightarrow$ more accurate forecasts.
- Frequent model updates may produce structural model changes $\Rightarrow$ forecasts can shift.

Key questions

- Can we maintain high accuracy while producing stable forecasts?
- Is frequent retraining necessary?

Experimental Design

Datasets & Models

Dataset	Frequency	N. Series	Test Set	Horizon
M5	Daily	30,490	364	28

Machine Learning	Deep Learning
Linear Regression (LR)	MLP
XGBoost	TCN
LightGBM (LGBM)	NBEATSx
CatBoost	NHITS

Evaluation Strategy: Rolling Origin

- **Why?** Out-of-sample testing assesses generalization
- **How?**
  - Train/test split, preserving temporal order
  - Train model on initial training set
  - Forecast h steps ahead
  - Shift origin forward by p steps
  - Repeat until the test set is exhausted
- **Window types:**
  - Fixed window: Only the most recent n observations used
  - Expanding window: Include all historical data up to the current origin
- **In our study:** Expanding window with a step size $p = 1$

Evaluation Metrics

Accuracy

- **Point**: Root Mean Squared Scaled Error (RMSSE) (1)
- **Probabilistic**: Scaled Multi-Quantile Loss (SMQL) (2),(3)

Stability

- **Point**: Symmetric Mean Absolute Percentage Change (sMAPC) (4)
- **Probabilistic**: Symmetric Multi-Quantile Percentage Change (sMQPC) (5),(6)

Sustainability Metrics

- Computing Time (CT)
- Cost (\$) and Savings (%)

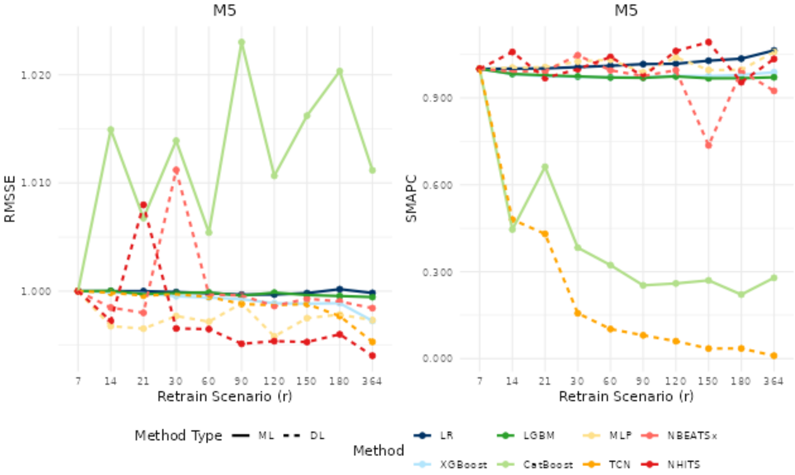
Simulation of the savings of each retraining scenario for a large retailer (200,000 SKUs and 5,000 stores), assuming some standard costs for computing services (\$3.5/hour).

Retraining Scenarios

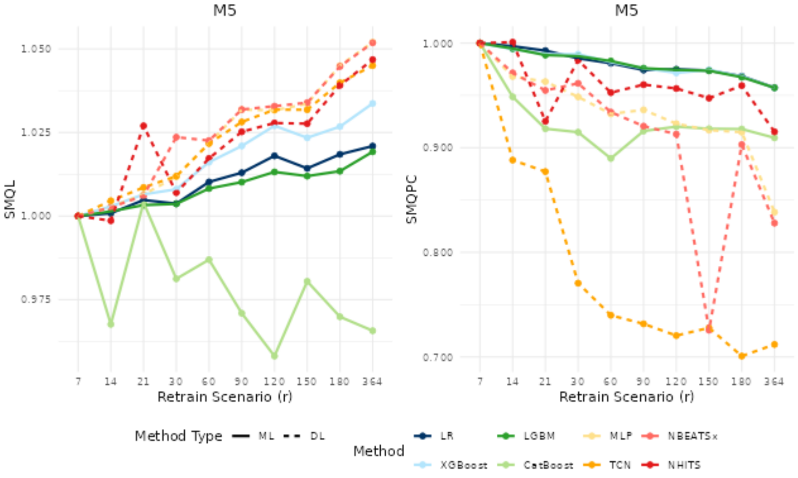
- **Retrain scenario (r)**: the interval at which the model is re-trained or updated.
- **Scenarios based on data frequency:**
  - Daily: $r = \{7, 14, 21, 30, 60, 90, 120, 150, 180, 364\}$
- **Three strategies:**
  - Continuous retraining, $r = 7$, baseline scenario
  - No retraining, $r = 364$
  - Periodic retraining, $7 < r < 364$

Empirical Results

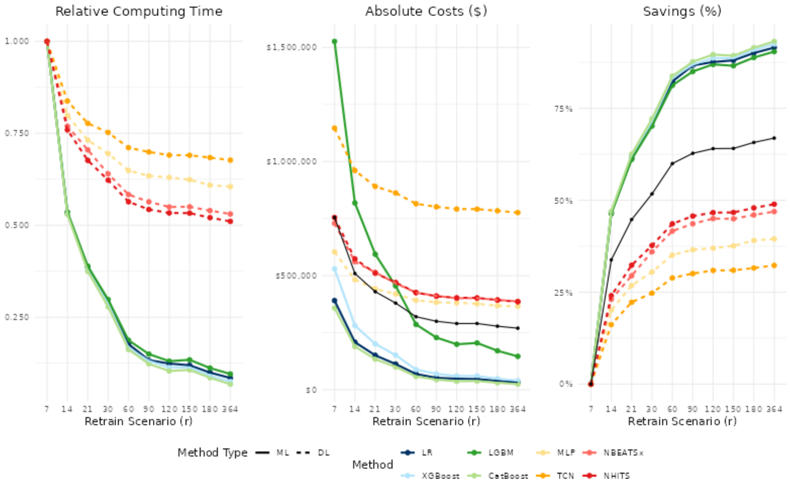
Point Forecasting



Probabilistic Forecasting



Sustainability



Conclusions

Key Results

Accuracy

- Point accuracy is stable across retraining scenarios
- Probabilistic accuracy slightly degrades with less frequent retraining but only for very low retraining frequencies

Stability

- Point stability is stable for lower retraining frequencies
- Probabilistic stability shows improvements for lower retraining frequencies

Sustainability

- Reducing the retraining frequency directly reduces the cost of producing forecasts up to 70% on average

Key Takeaways

- Highly frequent retraining may not be necessary
- Periodic retraining strategies allow for balancing accuracy and stability (especially in point forecasting)
- Reducing the retraining frequency supports more sustainable forecasting systems (lower costs of producing forecasts).

References

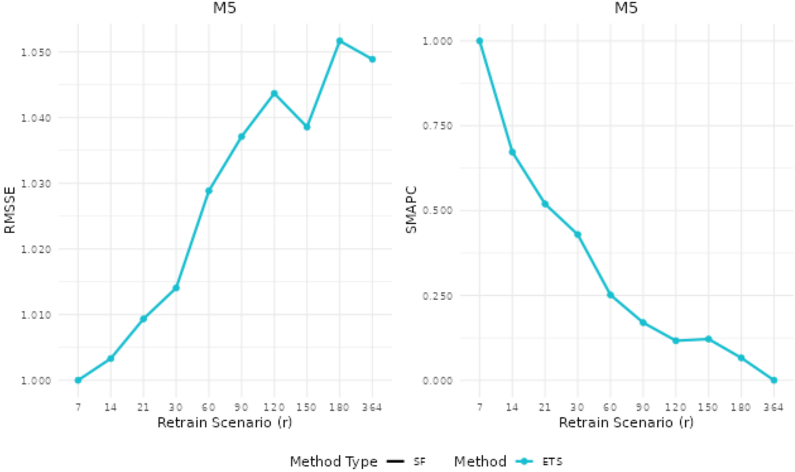
Zanotti, M. (2026). *Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems.* arXiv. (url)

Zanotti, M. (2025). *On the retraining frequency of global models in retail demand forecasting.* Machine Learning with Applications, 22, 100769. (url)

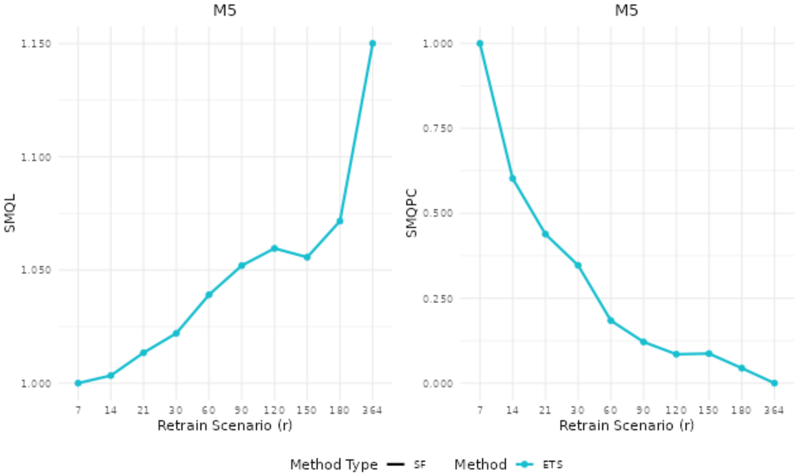
Thank you!

Appendix

Local Models - Point Forecasting



Local Models - Probabilistic Forecasting



Accuracy Metrics - Math

$$\text{RMSSE} = \sqrt{\frac{\frac{1}{h}\sum_{t=n+1}^{n+h}(y_t - \hat{y}_t)^2}{\frac{1}{n-s}\sum_{t=s+1}^{n}(y_t - y_{t-s})^2}} \tag{1}$$

$$\text{SQL}(\alpha) = \frac{\frac{1}{h}\sum_{t=n+1}^{n+h}\left(\alpha\cdot(y_t-\hat{q}_t^{(\alpha)})\cdot\mathbb{I}_{\hat{q}_t^{(\alpha)}\leq y_t}+(1-\alpha)\cdot(\hat{q}_t^{(\alpha)}-y_t)\cdot\mathbb{I}_{\hat{q}_t^{(\alpha)}>y_t}\right)}{\frac{1}{n-s}\sum_{t=s+1}^{n}|y_t-y_{t-s}|} \tag{2}$$

$$\text{SMQL} = \frac{1}{|\mathcal{Q}|}\sum_{\alpha\in\mathcal{Q}}\text{SQL}(\alpha) \tag{3}$$

Stability Metrics - Math

$$\text{sMAPC} = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{y}_{t,n} - \hat{y}_{t,n-1}|}{|\hat{y}_{t,n}| - |\hat{y}_{t,n-1}|} \tag{4}$$

$$\text{sQPC}(\alpha) = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{q}_{t,n}^{(\alpha)} - \hat{q}_{t,n-1}^{(\alpha)}|}{|\hat{q}_{t,n}^{(\alpha)}| - |\hat{q}_{t,n-1}^{(\alpha)}|}. \tag{5}$$

$$\text{sMQPC} = \frac{1}{|\mathcal{Q}|} \sum_{\alpha \in \mathcal{Q}} \text{sQPC}(\alpha), \tag{6}$$

Symmetric Multi-Quantile Percentage Change

The idea of sQPC is to mirror the sMAPC, but using quantile forecasts at different origins for some specific level α.

$$\mathrm{sQPC}(\alpha) = \frac{200}{h-1} \sum_{t=n+1}^{n+h-1} \frac{|\hat{q}_{t,n}^{(\alpha)} - \hat{q}_{t,n-1}^{(\alpha)}|}{|\hat{q}_{t,n}^{(\alpha)}| - |\hat{q}_{t,n-1}^{(\alpha)}|} \tag{7}$$

The sMQPC simply averages the SQPC across the set of quantiles $\mathcal{Q}$, measuring overall probabilistic instability.

$$\mathrm{sMQPC} = \frac{1}{|\mathcal{Q}|} \sum_{\alpha \in \mathcal{Q}} \mathrm{sQPC}(\alpha), \tag{8}$$

Computing Setup

- Microsoft Azure NC6s v3 VM
- 6 vCPUs, 1 GPU, 112 GB RAM
- Nixtla's libraries: `statsforecast`, `mlforecast`, `neuralforecast`